

Knapsack 0/1 problem using Dynamic programming. Knapsack 0/1 ①

Given: ① Objects with weight and profit,
② Knapsack with capacity.

O/P: list of objects included in Knapsack that has maximised profit.

i/p: Obj (items) : 1 2 3 ... n.
weight: w_i : w_1 w_2 w_3 ... w_n .
profit : p_i : p_1 p_2 p_3 ... p_n .
Knapsack's capacity $C = M$.

O/P: option. soln $(x_1, \dots, x_n)$
where $x_i = 0/1$. s.t.

(total profit is max) $\sum_i x_i p_i$ is max and.

$\sum x_i w_i \leq M$. (capacity constraint)

Wypil: Finding optimal soln $OPT(n, M)$ → no of obj → cap.

If n th object is ~~not~~ a part of optimal soln, then $OPT(n, M) = OPT(n-1, M)$.

else // n th obj. is part of optim. soln.

$OPT(n, M) = p_n + OPT(n-1, M - w_n)$ → option. soln of remaining obj.
↳ profit of n th obj.

Recursive form

Knapsack 0/1 (2)

$$OPT(i, c) = \begin{cases} OPT(i-1, c) & \text{if } w_i > c \\ \max \begin{cases} p_i + OPT(i-1, c-w_i) \\ OPT(i-1, c) \end{cases} & \text{else:} \end{cases}$$

(wt of obj > cap)
so obj i is excluded

// profit with obj i
// profit without including obj i

~~$i-1$~~
max. $\left\{ \begin{array}{l} \text{profit with obj } i \\ \text{profit without obj } i \end{array} \right.$

rate

$$i = n, n-1, \dots, 1$$

$$c = m, m-1, \dots, 1$$

Solution using dynamic programming

Notation: $M(i, j)$ = sum profit obtained from object 1, upto obj i , and capacity of knapsack = j .

Construct a matrix $M(\overbrace{0, \dots, n}^{\text{objects}}; \overbrace{0, \dots, m}^{\text{capacity}})$

- ① Initialize $M(0, c) = 0 \quad \forall c = 0, 1, 2, \dots, m$.
row 0 // no obj is included $\therefore$ profit = 0 for any cap = 0, $\dots$, m .
- ② for $i = 1, \dots, n$ // row 1, $\dots$, n , i.e., objects: 1, $\dots$, n .
{ for $c = 0$ to m // capacity is varied from 0 to m .

Knapsack 0/1 (3)

{ if ($w_i > c$)

// wt of obj-i' $>$ cap c
 w_i

$M(i, c) = M(i-1, c)$ // obj-i' can't be included

else // it obj-i' can be included

profit = max of { profit with obj-i',
profit without obj-i' }

$M(i, c) = \max \left\{ \begin{array}{l} M(i-1, c) \quad // \text{same profit without obj-i'} \\ p_i + M(i-1, c-w_i) \quad // \text{total profit with obj-i'} \end{array} \right.$

}

} Note: $M(n, M) \Rightarrow$ opt. soln i.e., max profit.
(last row, last column)

prob: solve knapsack prob for the ^{following} instance.

Objects: A B C

Weight w_i : 1 2 3

capacity = 5

profit p_i : 3 2 1

fnm. $M(i, j) =$ max. profit up to Object 'i' in a knapsack of cap = 'j'.

Knapsack 0/1 (4)

Profit P_i	Wt W_i	Object	cap	0	1	2	3	4	5
—	—	A	0	0	0	0	0	0	0
3	1	B	1	0	3	3	3	3	3
2	2	B	2	0	3	3	5	5	5
1	3	C	3	0	3	3	5	5	5

cap of knapsack giving

init. to 0 // no obj in bag

recursion $m(i, c) = \begin{cases} m(i-1, c) \\ P_i + m(i-1, c-w_i) \end{cases}$

option for including max. profit

row for $i=0$ to 3

col for $c=0$ to 5

cap of knapsack = 0 so profit = 0

if $w_i > c$ $m(i, c) = m(i-1, c)$
 else $m(i, c) = \max \{ m(i-1, c), P_i + m(i-1, c-w_i) \}$

which object is included/excluded?

From $m(i, j)$ matrix, $m(3, 5) = 5$ is the max. profit.

max profit with obj '3'

Now max. profit excluding obj '3' = $m(2, 5) = 5$

Since there is no diff between $m(3, 5)$ & $m(2, 5)$

$m(3, 5) = 5$, including object '3' does not increase profit. $\therefore$ exclude obj '3'. i.e., c is not taken.

Since $m(2,5) \geq m(1,5)$

including object 2 increases profit

$\therefore$ Take object 2 i.e., B ✓

Now $m(2,5)$ is obtained from $P_2 + m(1,5-2) = P_2 + m(1,3)$
 $= 2 + 3$

Go to $m(1,3) = 3$ // with obj 1

also $m(0,3) = 0$ // with obj 1

Since $m(1,3) > m(0,3)$, including obj 1 increases profit.

$\therefore$ Take object 1 i.e., A ✓

Since we reached limit object, construct soln.

A ✓, B ✓, C ✗ $\therefore x_1=1, x_2=1, x_3=0$

i.e., $(1, 1, 0)$

OR: Alternate explanation: $OPT(3,5) = m(3,5) = 5$

$m(3,5)$ is obtained from $m(2,5)$ i.e., $m(3,5) = m(2,5)$
 So don't include obj 3.

Now $m(2,5)$ is obtained from $P_2 + m(1,3)$. So include object 2.

$m(1,3)$ is obtained from $P_1 + m(0,2)$. So include object 1.

$\therefore$ Option soln is $(x_1, x_2, x_3) = (1, 1, 0)$ ✓

max profit is $P_1 + P_2 = 3 + 2 = 5$

Prob: Using DP solve following
Knapack 01.

Object: A B C D

Weight w_i : 6 2 5 3

Profit p_i : 30 14 32 12

Capacity of Knapack: 8

Soln

p_i	w_i	obj	c	0	1	2	3	4	5	6	7	8
30	6	A	1	0	0	0	0	0	0	0	0	0
14	2	B	2	0	0	14	14	14	14	30	30	44
32	5	C	3	0	0	14	14	14	32	32	46	46
12	3	D	4	0	0	14	14	14	32	32	46	46

init to 0

option soln

row 1: $\max \{m(0, c), p_1 + m(0, c - w_1)\}$

row 2: $\max \{m(1, c), p_2 + m(1, c - w_2)\}$

row 3: $\max \{m(2, c), p_3 + m(2, c - w_3)\}$

row 4: $\max \{m(3, c), p_4 + m(3, c - w_4)\}$

$m(4, 8) = 46$ option soln.

$m(4, 8)$ is from $m(3, 8) \Rightarrow$ obj 4 not included.

$m(3, 8)$ is from $p_3 + m(2, 3) \Rightarrow$ obj 3 is included.

$m(2, 3)$ is from $p_2 + m(1, 1) = 14 + 0 \Rightarrow$ obj 2 is included.

$m(1, 1)$ is from $m(0, 1) \Rightarrow$ don't include obj 1.
Soln is $(n_1, n_2, n_3, n_4) = (0, 1, 1, 0)$